NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 24/01/2026
Time: 3 Hrs | GTM – 24 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	B	C	D	C	A	B	D	C	C
11	12	13	14	15	16	17	18	19	20
C	C	B	C	C	A	A	B	B	B
21	22	23	24	25					
12	6	400	12	37					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	C	B	D	A	C	C	B	B	A
36	37	38	39	40	41	42	43	44	45
D	A	C	B	B	A	A	B	D	A
46	47	48	49	50					
75	56	99	3	4					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	C	D	B	D	D	C	B	B	C
61	62	63	64	65	66	67	68	69	70
C	B	C	A	D	C	A	B	D	A
71	72	73	74	75					
3	403	5	19	101					

Rankers Academy JEE

or More Join: @IITJEE_Advanc

HINTS

PHYSICS

1. Electric field inside closed conductor (cavity) is zero

2. $\textbf{density } \rho = \frac{M}{V} = \frac{A(1amu)}{\left(\frac{4}{3}\pi R^3\right)} = \frac{A(1amu)}{\frac{4}{3}\pi A R_0^3}$

   $\rho \rightarrow$ Const independent on mass number

   $R = R_0 A^{\frac{1}{3}} \qquad R \alpha A^{\frac{1}{3}}$

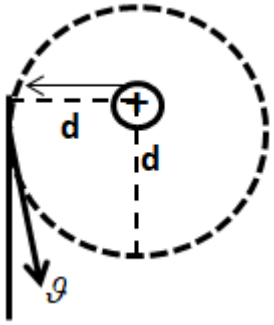
3. $I = \frac{B_0^2}{2\mu_0} C$

   $B_0 = \sqrt{\frac{2I\mu_0}{C}} \qquad B_0 \rightarrow$ Magnetic filed

4. $C = \frac{\varepsilon_0 A}{d} \qquad L = \frac{\mu_0 NA}{l} \qquad \frac{C}{L} \alpha \frac{\varepsilon_0}{\mu_0} = \frac{\tau/R}{\tau R} = \frac{1}{R^2}$

   $\frac{1}{R} = \sqrt{\frac{\varepsilon_0}{\mu_0}} \rightarrow$ conductance

5. $\frac{mv^2}{d} = \frac{e^2}{4\pi\varepsilon_0 d^2}$



   $KE = \frac{e^2}{8\pi\varepsilon_0 d}$

   But $KE_{max} = \frac{hc}{\lambda} - W \qquad \frac{hc}{\lambda} = \frac{e^2}{8\pi\varepsilon_0 d} + W$

   $\frac{hc}{\lambda} = \frac{e^2 + 8\pi\varepsilon_0 dW}{8\pi\varepsilon_0 d} \qquad \lambda = \frac{8\pi\varepsilon_0 dhc}{e^2 + 8\pi\varepsilon_0 dW}$

6. $\frac{dA}{dt} = \frac{L}{2m}$

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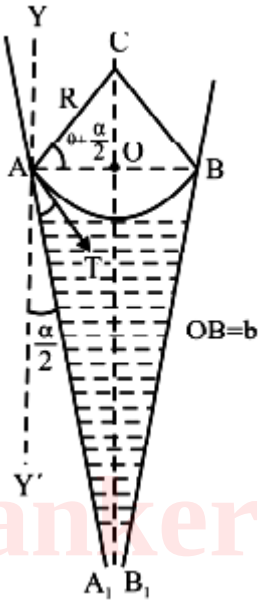
7. Internal Energy $U=\frac{1}{2}fnRT$

for O_2 → diatomic degrees of freedom f=5

for Ar →monoatomic gas degrees of freedom f=3

$$U = U_1 + U_2 = \frac{2\times 5RT}{2} + \frac{4\times 3RT}{2} = 11RT$$

8.



$$\cos\left[\theta+\frac{\alpha}{2}\right]=\frac{b}{R}$$

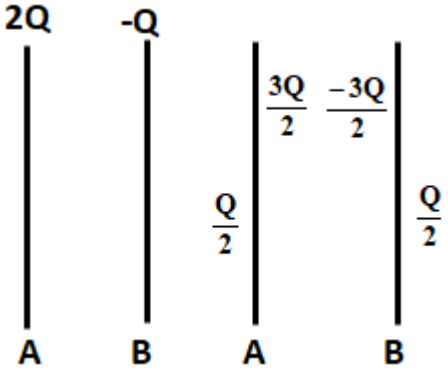
Capillary height 'h' $=\frac{2s}{R\rho g}=\frac{2s}{b\rho g}\cos\left[\theta+\frac{\alpha}{2}\right]$

9. $V=\frac{u}{2}$ $u=-30cm$ $v=-15cm$ $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$

$$\frac{1}{f}=-\frac{1}{15}-\frac{1}{30}=-\frac{1}{10} \quad f=-10cm$$

10. Dipole moment P = q (a)

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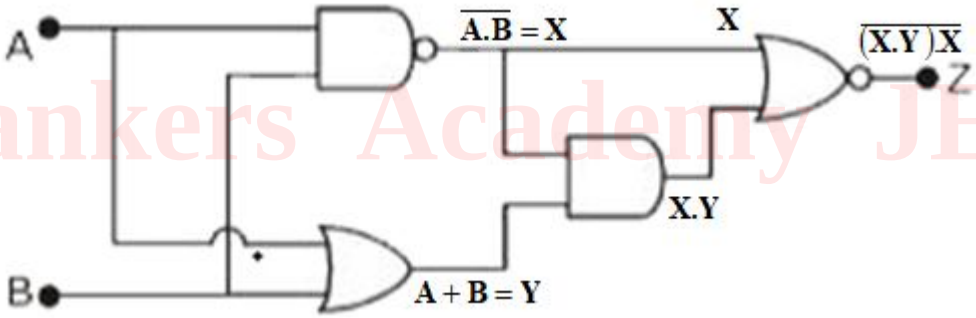
$$P = 5\times10^{-6}\times2\times10^{-7} = 10^{-12}$$

$$E_{ln} = \frac{v}{d} = \frac{3Q}{2cd}$$

$$E = \frac{3\times2\times10^{-6}}{2\times5\times10^{-6}\times3\times10^{-4}} = \frac{1}{5}\times10^{4} = 2000$$

$$\tau = PE\sin\theta = 10^{-12}\times2\times10^{3}\times\frac{1}{2} = 10^{-9}\,Nm$$

11.



A	B	X	Y	X.Y	$\overline{(X.Y)+X}$
1	0	1	1	1	0
0	0	1	0	0	0
1	1	0	1	0	1
0	1	1	1	1	0

12. Torque $\rightarrow \tau = r\times F \rightarrow ML^2T^{-2}$

Angular Momentum $\rightarrow L = mvr \rightarrow ML^2T^{-1}$

Moment of Inertia $\rightarrow I = Mr^2 \rightarrow ML^2$

Radius of Gyration $\rightarrow K = \sqrt{\frac{I}{m}} \rightarrow L$

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13. $V = f\lambda \qquad 360 = 500\lambda$

$$\Delta\varphi = \frac{2\pi}{\lambda}\Delta x \qquad \frac{\pi}{3} = \frac{2\pi}{\lambda}\Delta x$$

$$\Delta x = \frac{\lambda}{6} = \frac{1}{6}\left[\frac{360}{500}\right] = 12\text{cm.}$$

14. Bending of light depends on optical density refered by refractive index of medium. Hence reason is not correct.

15. conceptual

16. $F = -\frac{du}{dr}$

17. conceptual

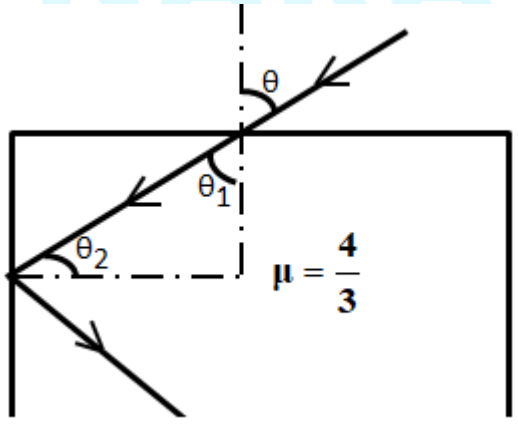
18. $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 1960}{9.8}} = 20\text{sec}$

$$AB = ut = 600\times\frac{5}{18}\times 20 = 3333.33\text{m} = 3.33\text{km}$$

19. $v = \frac{dy}{dt} = 5 + 16t \qquad a = \frac{dv}{dt} = 16$

$$F = ma = 0.5\times 16 = 8\text{N}$$

20.



$$1\sin\theta = \frac{4}{3}\sin\theta_1 \rightarrow (1)$$

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$$1\sin 90^0 = \frac{4}{3}\sin\theta_2$$

$$1 = \frac{4}{3}\sin[90-\theta_1]$$

$$1 = \frac{4}{3}\cos\theta_1 \quad \rightarrow (2)$$

$$(1)^2 + (2)^2 \qquad \sin^2\theta + 1 = \frac{16}{9}$$

$$\sin^2\theta = \frac{7}{9}$$

21. $Q_{in} = Q\left[1-\frac{1}{k}\right]$

$$8\times10^{-6} = Q\left[1-\frac{1}{3}\right]$$

$$8\times10^{-6} = \frac{2Q}{3}$$

$$Q = 12\times10^{-6}\,C$$

22. Power factor $\cos\varphi = \frac{R}{2} = \frac{R}{\sqrt{R^2+(X_L-X_L)^2}}$

$$\cos\varphi = \frac{60}{\sqrt{60^2+(100-20)^2}} = \frac{60}{100} = \frac{3}{5}$$

$$10\times\frac{3}{5} = 6$$

23. $\frac{dQ}{dt} = H = \frac{KA\,d\theta}{l}$

$$H_1 = 40 = \frac{K_S A\,d\theta}{l} \qquad \frac{3}{K_S} = \frac{1}{K_{Al}} + \frac{1}{K_{cu}} + \frac{1}{K_{Al}}$$

$$H_2 = \frac{K_\rho A\,d\theta}{l} \qquad \frac{3}{K_S} = \frac{5}{400} \qquad K_S = 240$$

$$\frac{H_2}{40} = \frac{\frac{800}{3}(3A)d\theta}{\frac{l}{240(A)d\theta}{3l}}$$

$$K_P = \frac{K_{Al}+K_{cu}+K_{Al}}{3}$$

$$\frac{H_2}{40} = \frac{800}{80} \qquad K_P = \frac{800}{3}$$

$$H_2 = 400\,W$$

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24. $\varphi = \bar{E}.\bar{A} = \left[3\hat{i} + 2\hat{j} + 6\hat{k}\right].2\hat{k} = 12$

25. $I_{Cylinder} = \dfrac{\left(\left(\pi R^2\right)3R\rho\right)R^2}{2}$

$I_{Sphere} = \dfrac{2}{5}\left[\dfrac{2}{3}\pi R^3\rho\right]R^2$

$I_{remaining} = I_{Cylinder} - I_{Sphere}$

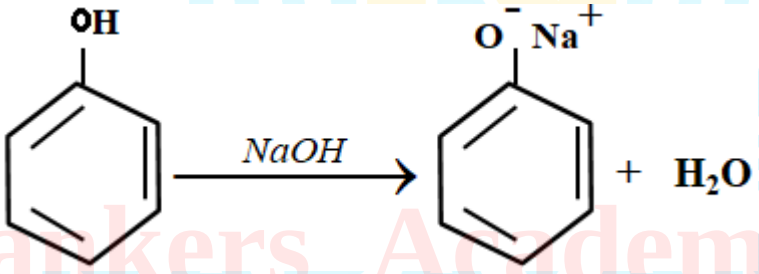
$= \left[\dfrac{3}{2} - \dfrac{4}{15}\right]\pi R^5\rho$

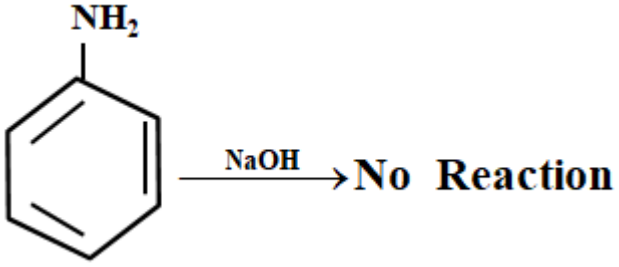
$= \dfrac{37}{30}\pi R^5\rho$

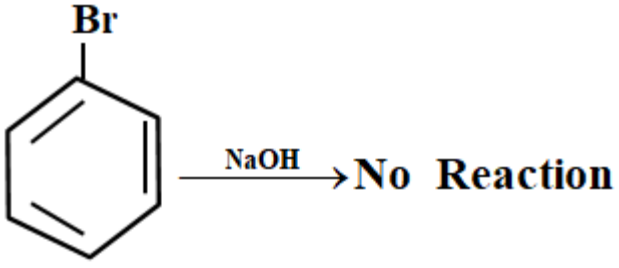
CHEMISTRY

26. NCERT

27.







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28. $\Delta H = L.E + H.E_{Cation} + H.E_{Anion}$

$d = L.E + (-a) + (-b)$

$L.E = d + a + b$

29. $A_{(g)} \rightarrow 2B_{(g)} + C_{(g)}$

t = 0	P^0	0	0
t = t	P^0-x	2x	x
$t = \infty$	0	$2P^0$	P^0

$P_t = P^0 - x + 2x + x$

$P_t = P^0 + 2x$

$x = \dfrac{P_t - P^0}{2} \rightarrow (1)$

$P_\infty = 2P^0 + P^0$

$P_\infty = 3P^0$

$\therefore P^0 = \dfrac{P_\infty}{3} \rightarrow (2)$

2 in 1

$\xrightarrow{(1)} x = \dfrac{P_t - \left(\frac{P_\infty}{3}\right)}{2}$

$\therefore K = \dfrac{1}{t} \ln \dfrac{\left(\frac{P_\infty}{3}\right)}{\left(\frac{P_\infty}{3} - \frac{(3P_t - P_\infty)}{6}\right)}$

$x = \dfrac{3P_t - P_\infty}{6}$

$\therefore K = \dfrac{1}{t} \ln \dfrac{2P_\infty}{3(P_\infty - P_t)}$

30 $CH_3 - \overset{O}{\overset{\|}{C}} - CH_2 - \underset{CH_3}{CH} - CH_3 \xrightarrow{2,4\ DNP}$ gives + ve Test

$CH_3 - \overset{O}{\overset{\|}{C}} - CH_2 - \underset{CH_3}{CH} - CH_3 \xrightarrow{clemensen,Reduction} CH_3 - CH_2 - CH_2 - \underset{CH_3}{CH} - CH_3$

(2 – methyl pentane)

31. $Cl^-_{(18e^-)} > Cl_{(17e^-)}$

Inert gas configuration

32. Less PK_a more acidic

33. $\frac{1}{2} S_{8(Rhombic)} + O_{2(g)} \rightarrow So_{2(g)}$

(one mole of So_2 forms form its elements in standard states)

34. In Liquid Phase

$P_T = P^0 X_A + P_B^0 X_B$

$P_T = 7 \times 10^3 \times 0.4 + 12 \times 10^3 \times 0.6$

In Vapour Phase

$P_A = P_T . Y_A$

$Y_A = \dfrac{P_A}{P_T} = \dfrac{P_A^0 X_A}{P_T}$

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$P_T = 2.8\times10^3 + 7.2\times10^3$

$Y_A = \frac{7\times10^3\times0.4}{10^4} = \frac{2.8\times10^3}{10^4}$

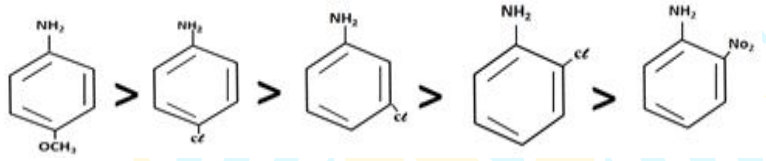
$P_T = 10^4 P_a$

$Y_A = 0.28\ ;\ Y_B = 1 - 0.28$

$Y_B = 0.72$

35. $K_2CrO_4, CrO_3, KMnO_4$ coloured due to charge transfer phenomenon

36.



Pk_a :5.29 Pk_a :3.98 rk_a : 3.34 Pk_a :2.64 Pk_a : – 0.29

→More PK_a More Basic

37. Primary valency = oxidation state
Secondary Valency = Coordination Number

		Primary valency	Secondary Valency
A	$[Co(NH_3)_5Cl]Cl_2$	3	6
B	$[Cu(en)_2]SO_4$	2	4
C	$[Pt(NH_3)_5Cl]Cl_3$	4	6
D	$[Mg(EDTA)]^{-2}$	2	6

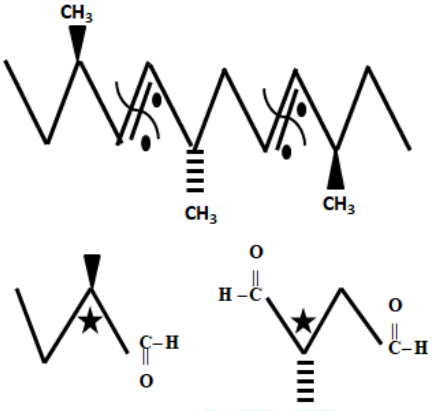
38. A) Ethanol + water - Minimum B.P Azeotrope
B) n- Hexane + n –Heptane – Ideal solution; $\Delta H_{Mix} = 0$
C) $HNO_3 + H_2O$ – Maximum B.P Azeotrope $(68\%\ HNO_3 + 32\%\ H_2O)$
D) $HCl + H_2O$ – Maximum B.P Azeotrope

39. $CH_3^+ - Sp^2$
$NO_3^- - Sp^2$
$I_2Cl_6 - sp^3d^2$

40. Benzaldehyde cannot give Fehiling's test

41. 2-only

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42. conceptual
43. B) larger exchange energy
    d) Symmetrical distribution of e^-
44. A) Tl^{+1} is more stable due to inert pair effect
    B) Tl^{+3} is a powerful oxidising agent
    C) Al^{+3} and Tl^{+1} are highly stable
    D) Tl^{+1} is most stable due to inert pair effect
45. I) Cu, Ag deposited at cathode
    II) Na, Mg, and K instead H_2 evolved at cathode.
46. $\mathbf{Ph = pk_a + \log\frac{[salt]}{Acid}}$ $\begin{pmatrix}\mathbf{V = sodium\ Benzoate}\\ \mathbf{225 - V = Benzoic\ acid}\end{pmatrix}$

$$\mathbf{4.5 = 4.2 + \log\left(\frac{1M \times V}{1M \times (225 - V)}\right)}$$

$$\mathbf{0.3 = \log\left(\frac{V}{225 - V}\right)}$$

$$\mathbf{\log 2 = \log\left(\frac{V}{225 - V}\right)}$$

$$\mathbf{2(225 - V) = V}$$

$$\mathbf{450 - 2V = V}$$

$$\mathbf{3V = 450 \Rightarrow V = \frac{450}{3} = 150}$$

Vol of benzoicacid = 225-V

=225-150=75ml

47. $0.2 \times 2 \times V_{ml} = 0.1 \times 1 \times 80$

$V_{ml} = 20$

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Acid Remaining volume = 100-20=80ml N=M×n-f

$$\% N = \frac{1.4 \times N \times V}{W.O.C} = \frac{1.4 \times 0.2 \times 2 \times 80}{0.8} = \frac{14 \times 2 \times 2 \times 8}{8} = 56\%$$

48. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

44g 160g 4×18g

70kg 220kg

(O_2 – Limiting Reagent)

$\therefore 160g \rightarrow 4 \times 18g$

$220kg \rightarrow x$

$$x = \frac{220 \times 10^3 \times 4 \times 18}{160} = 99 \times 10^3 g$$

$$d = \frac{m}{v}$$

$$1 = \frac{99 \times 10^3 g}{v} \Rightarrow v = 99 \times 10^3 ml = 99lit$$

49. 1) $[Zn(NH_3)_4]^{+2} - Zn^{+2} = [Ar]4s^0 3d^{10} - 0$ unpaired e^- (Diamagnetic)

2) $[Ni(CO)_4] - Ni^0 = [Ar]4s^0 3d^{10} - 0$ unpaired e^- (Diamagnetic)

3) $[Ni(CN)_4]^{+2} - Ni^{+2} = [Ar]4s^0 3d^8 - 0$ unpaired e^- (Diamagnetic)

4) $[Fe(CN)_6]^{-3} - Fe^{+3} = [Ar]4s^0 3d^5 - 1$ paired e^- (Paramagnetic)

5) $[Cr(NH_3)_6]^{+3} - Cr^{+3} = [Ar]4s^0 3d^3 - 3$ paired e^- (Paramagnetic)

6) $[Fe(H_2O)_5 NO]So_4 - Fe^{+1} = [Ar]4s^0 3d^7 - 3$ unpaired e^- (Paramagnetic)

7) $[CoF_6]^{-4} - Co^{+2} = [Ar]4s^0 3d^7 - 3$ unpaired e^- (Paramagnetic)

8) $[Fe(H_2O)_6]^{+3} - Fe^{+3} = [Ar]4s^0 3d^5 - 5$ unpaired e^- (Paramagnetic)

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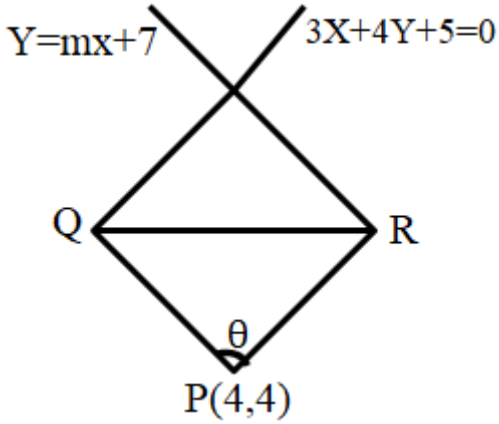
50. $A = CH_3 - \overset{\overset{OH}{|}}{CH} - CH_3$

$B = CH_3 - CH = CH_2$

$C = CH_3 - CH_2 - CH_2 - OH$

MATHEMATICS

51. $A = \frac{1}{2} \times PQ.PR.\sin\theta$



$= \frac{1}{2} \times \frac{33}{5} \times \frac{|4m+3|}{\sqrt{1+m^2}} \sin\theta$

$= \frac{33}{2}\sin^2\theta \quad \left[\sin\theta = \frac{|4m+3|}{5\sqrt{1+m^2}}\right]$

It is maximum for $\theta = \frac{\pi}{2}$ and $m = \frac{4}{3}$

52. $\vec{b_1}.\vec{a} = \vec{b}.\vec{a} - \frac{\vec{b_1}.\vec{a}}{|\vec{a}|2}.\left(\vec{a}.\vec{a}\right) = 0$

$\vec{a}$ and $\vec{b_1}$ are orthogonal

$\vec{c_1}.\vec{b_1} = \vec{c}.\vec{b_1} - 0 - \frac{\vec{b_1}.\vec{c}}{|\vec{b_1}|}.\left(\vec{b_1}.\vec{b_1}\right) = 0$

$\vec{b_1}.\vec{c_1} = 0$

53. $A: -5 \le x-1 \le 5 \;;\; -6 \le y-6 \le 6$

$-4 \le x \le 6 \quad 0 \le y \le 12$

$B: \frac{(x-2)^2}{9} + \frac{(y-6)^2}{16} \le 1$

$\rightarrow B < A$

54. $-3 < \frac{x^2+ax-2}{x^2+x+1} < 2$

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Case - 1:- $4x^2+(a+3)x+1>0$

$(a+3)^2-16<0$

$(a-1)(a+7)<0$ ---------- (1)

Case - 2:- $\frac{x^2+ax-2}{x^2+x+1}<2$

$x^2+x(2-a)+4>0$

$(2-a)^2-16<0$

$(a+2)(a-6)<0$ ---------- (2)

From (1) and (2), $a\in(-2,1)$

55. Box – I: 3 – White, 2 – Red; Box-II: 1 - White

E_1 : Head

E_2 : Tail

A: white ball drawn from Box – II

$P(E_1)=P(E_2)=\frac{1}{2}$

$P\left(\frac{A}{E_1}\right)=\frac{3C_1}{5C_1}\times\frac{2C_1}{2C_1}+\frac{2C_1}{5C_1}\times\frac{1C_1}{2C_1}=\frac{4}{5}$

$P\left(\frac{A}{E_2}\right)=\frac{3C_2}{5C_2}\times\frac{3C_1}{3C_1}+\frac{3C_1+2C_1}{5C_2}\times\frac{2C_1}{3C_1}+\frac{2C_2}{3C_2}\times\frac{1C_1}{3C_1}=\frac{11}{15}$

$P\left(\frac{E_1}{A}\right)=\frac{\frac{1}{2}\times\frac{4}{5}}{\frac{1}{2}\times\frac{4}{5}+\frac{1}{2}\times\frac{11}{15}}=\frac{12}{23}$

56. $0\le\frac{1+3p}{4}\le1\Rightarrow\frac{-1}{3}\le p\le1$

$0\le\frac{1-p}{4}\le1\Rightarrow-3\le p\le1$ $\Rightarrow-\frac{1}{3}\le P\le\frac{1}{4}$

$0\le\frac{1+2p}{4}\le1\Rightarrow\frac{-1}{3}\le p\le\frac{3}{2}$

$0\le\frac{1+4p}{4}\le1\Rightarrow\frac{-3}{4}\le p\le\frac{1}{4}$

Mean $=\frac{2-9p}{4}$ $\Rightarrow-\frac{1}{16}\le\frac{2-9p}{4}\le\frac{5}{4}$

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57.



$$\text{Area} = \frac{3}{2}\ln\left(\frac{4}{3}\right) = \ln\left(\frac{8}{3\sqrt{3}}\right)$$

58. $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots\ldots\ldots + a_6x^6$

$$\lim_{x\to 0}\left(1+\frac{f(x)}{x^3}\right)^{\frac{1}{x}} = e^2 \Rightarrow e^2 \Rightarrow a_0 = a_1 = a_2 = a_3 = 0$$

$$e^{\lim_{x\to 0} a_4 + a_5x + a_6x^2} = e^2 \Rightarrow a_4 = 2$$

$$f(x) = 2x^4 + a_5x + a_6x^6$$

$$f^1(1) = 0,\ f^1(2) = 0 \Rightarrow a_5 = \frac{-12}{5}, a_6 = \frac{2}{3}$$

$$\therefore\ f(x) = 2x^4 - \frac{12}{5}x^5 + \frac{2}{3}x^6$$

59. $2^{\sec^2 x} \geq 2$ *and* $\left(y^y - y^2 + \frac{1}{2}\right) \geq \frac{1}{4}$

$$\therefore\ 2^{\sec^2 x}\sqrt{\frac{1}{2} - y^2 + y^4} \leq 1$$

$$\Rightarrow 2^{\sec^2 x} = 2 \quad \textit{and} \quad y^y - y^2 + \frac{1}{2} = \frac{1}{4}$$

$$\Rightarrow \sec^2 = 1 \quad \textit{and} \quad y^2 = \frac{1}{2}$$

$$\Rightarrow n = 0,\ \pi, 2\pi, 3\pi \quad \textit{and} \quad y = \pm\frac{1}{\sqrt{2}}$$

$\therefore$ 8 pairs of (x, y) are possible

60. $sum = \left(a_1^2 - a_2^2\right) + \left(a_3^2 - a_4^2\right) + \ldots\ldots\ldots + \left(a_{49}^2 - a_{50}^2\right)$

$$= (-D)\left(a_1 + a_2 + a_3 + a_4 + \ldots\ldots\ldots + a_{49} + a_{50}\right)$$

$$= (-D)|25|\left(a_1 + a_{50}\right)$$

$$= \frac{(-D)}{a_1 - a_{50}}(25)\left(a_1^2 - a_{50}^2\right)$$

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$$=\frac{1}{49}(25)\left(a_1{}^2-a_{50}{}^2\right)$$

$$\Rightarrow n=8$$

61 Let $Z=x+iy$

$$x+iy=\frac{3i-t}{2+it}$$

$$\Rightarrow (x+iy)(2+it)=3i-t$$

Compare real and imaginary parts,

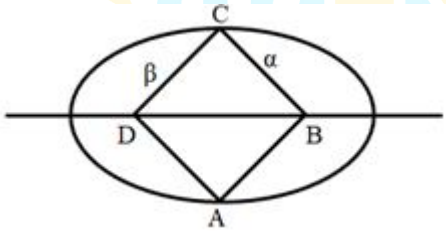
$2x-yt=-t$ *and* $2y+tx=3$

Eliminate t, we get

$$x^2+y^2-\frac{5y}{2}+\frac{3}{2}=0$$

Centre $c\left(0,\frac{5}{4}\right)$, $r=\frac{1}{4}$

62



$$\alpha\beta=200$$

$$\pi ab=200\pi$$

$$\alpha^2+\beta^2=4a^2e^2\Rightarrow\alpha^2+\beta^2=4\left(a^2-b^2\right)$$

$$=(\alpha+\beta)^2-2\alpha\beta=4\left(a^2-b^2\right)$$

$$=(2a)^2-2\alpha\beta=4a^2-4b^2$$

$$=b^2=100\Rightarrow b=10\Rightarrow a=20$$

Perimeter $=2(\alpha+\beta)=4a=80$

63. $xydx+x^2dy+y^4dx-xy^3dy=0$

$$\Rightarrow xd(xy)-y^3(xdy-ydx)=0$$

$$\Rightarrow xd(xy)-y^3.x^2d\left(\frac{y}{x}\right)=0$$

$$\Rightarrow\frac{d(xy)}{(xy)^2}-\left(\frac{y}{x}\right)d\left(\frac{y}{x}\right)^2=0$$

$$\Rightarrow\frac{-1}{xy}-\frac{1}{2}\left(\frac{y}{x}\right)^2=c$$

$y(4)=-2\Rightarrow c=0$ $\quad\therefore y^3+2x=0$

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$$\int_{\frac{-1}{2}}^{\frac{1}{2}}[f(n)]dx=\int_{\frac{-1}{2}}^{0}0dx+\int_{0}^{\frac{1}{2}}(-1)dx=\frac{-1}{2}$$

64. $A(s+1,-s-1,s-1)$

$B(-t-2,-t+1,t+2)$

DR's of $AB: s+t+3,\ t-s-2,\ s-t-3$

$AB\perp L_1 \Rightarrow 3s-t+2=0$

$AB\perp L_2 \Rightarrow s-3t-4=0$

$s=\frac{-5}{4},\ t=\frac{-7}{4}$

$$AB:\frac{x+\frac{1}{2}}{0}=\frac{y-\frac{1}{4}}{\frac{-5}{2}}=\frac{z+\frac{9}{4}}{\frac{-5}{2}}$$

65. $x_1\ A\ x_2\ B\ x_3\ C\ x_4\ D\ x_5$

$x_1+x_2+x_3+x_4+x_5=8$

$x_1\geq 0, x_2, x_3, x_4>2,\ x_5\geq 0$

P = no of arrangements = $6c_4\times 4!=360$

$x_1\ A\ x_2\ CD\ x_3$

$x_1+x_2+x_3=8$

$x_1\geq 0,\quad x_2\geq 1,\quad x_3\geq 0$

q = no of arrangements = $9c_2\times 4!=864$

$p+q=1224$

66. $A(5\lambda-3,\ 2\lambda+1,\quad 3\lambda-4)$

$B(\mu+1,\ 3\mu-2,\quad 2-2\mu)$

DR's of AB : $5\lambda-\mu-4,\ 2\lambda-3\mu+3,\ 3\lambda+2\mu-6$

$$\frac{5\lambda-\mu-4}{0}=\frac{2\lambda-3\mu+3}{2}=\frac{3\lambda+2\mu-6}{-1}$$

$\Rightarrow \lambda=1,\quad \mu=1$

$\therefore AB=\sqrt{5}$

67. For $x<-1,\quad x=-2$

For $-1\leq x<0$, no Solution

For $0\leq x<1$, No Solution

$1\leq x<2$, No Solution

$x>2\quad x\in[2,\infty)$

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$\therefore\ x \in \{-2\} \cup [2, \infty)$

68. $CC_1 = r + r_1$

$CC_2 = r + r_2$

$CC_1 - CC_2 = r_1 - r_2 = 2a$

$C_1C_2 = r_1 + r_2 = 2ae$

$e = \dfrac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$

69. $\begin{vmatrix} t-1 & 3t+1 & 2t \\ t-1 & 4t-2 & t+3 \\ 2 & 3t+1 & 3(t-1) \end{vmatrix} = 0$

$R_1 : R_1 - R_2;\ R_2 : R_2 - R_3$

$\begin{vmatrix} 0 & -t+3 & t-3 \\ t-3 & t-3 & -2t+6 \\ 2 & 3t+1 & 3(t-1) \end{vmatrix} = 0$

$\Rightarrow 6t(t-3)^2 = 0 \qquad \Rightarrow t = 0, 3$

70. $\dfrac{a\left(1-r^{201}\right)}{1-r} = 625$

$\sum_{i=1}^{201} \dfrac{1}{a_i} = \dfrac{\frac{1}{a}\left(\left(\frac{1}{r}\right)^{201} - 1\right)}{\left(\frac{1}{r} - 1\right)} = \dfrac{625}{\left(ar^{100}\right)^2} = 1$

71. $f(x) = \begin{cases} 1 - \cos\left(1 - \tan\left(\dfrac{\pi}{4} - x\right)\right), & x > 0 \\ \dfrac{\lambda \sin 4x}{x}, & x < 0 \end{cases}$

72. $\int x\left(x^{2009} + x^{803} + x^{401}\right)\left(2x^{1608} + 5x^{402} + 10\right)^{\frac{1}{402}} dx$

$\int \left(x^{2009} + x^{803} + x^{401}\right)\left(2x^{1608} + 5x^{402} + 10\right)^{\frac{1}{402}} dx$

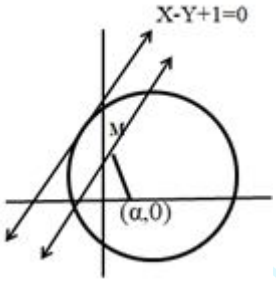
$Sub\ \ t = 2x^{2010} + 5x^{804} + 10x^{402}$

$\alpha = 403$

73. Use Expansions $n = 5$

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74.



$r = \perp^r$ distance from $(\alpha, 0)$ to $x - y + 1 = 0$

$\Rightarrow 2r^2 = (\alpha + 1)^2$ ..... (1)

Now,

$$\frac{(-3\alpha + 0 - 1)^2}{(\sqrt{13})^2} + \left(\frac{2}{\sqrt{13}}\right)^2 = r^2$$

$\Rightarrow (3\alpha + 1)^2 + 4 = 13r^2$ ..... (2)

From (1) and (2) $\alpha = \frac{-1}{5}, 3$

$\therefore\ r = 2\sqrt{2}$

ae =3 and $2a = 4\sqrt{2}$

$a^2 = 8$

$b^2 = 1$

75. $s = \sum_{r=0}^{100} (2r+1).(100C_r)^2$ ..... (1)

$s = \sum_{r=0}^{100} (2(100 - r) + 1).(100C_{100-r})^2$ ..... (2)

From (1) and (2),

$s = \frac{1}{2}\sum_{r=0}^{100} 202.(100C_r)^2$

$= 101 \times 200C_{100}$

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